NORTH SOUTH UNIVERSITY
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Spring 2026
EEE/ETE 111L.18 Lab | Analog Electronics Lab I

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Office Room: TBA
Office Hours: TBA

Lab Section: 18
Lab Timing: 2:40 PM – 5:50 PM
Room no: SAC 506

Course Description
This Lab course involves performing experiments based on the EEE/ETE 111 theory class.

Classes	Lab No	Title of Activities	Date
Class 1	Lab 01	Group Formation, Introductory Class and I-V Characteristics of Diode	31/1/2026
Class 2	Lab 02	Half Wave and Full Wave Rectifier Circuits	07/2/2026
Class 3	Lab 03	Clipper Circuits and Clamper Circuits	14/2/2026
Class 4	Lab 04	Zener Diode Characteristics and Applications	**28/3/2026**
Class 5		**Mid Exam**	**7/3/2026**
Class 6	Lab 05	The Output Characteristics of BJT for CE Configuration	14/3/2026
Class 7	Lab 06	The BJT Biasing Circuits	28/3/2026
Class 8	Lab 07	Study of Switching Characteristics of MOSFET	4/4/2026
Class 9		**Setup, Simulation Test and Viva**	**11/4/2026**
Class 10		**Final Exam**	**18/4/2026**

Tentative Marks Distribution

Attendance & Class Performance	15%
Lab Report	15%
Mid Exam	20%
Viva	10%
Setup and Simulation Test	15%
Final Exam	25%

Rules

- Arrive in class on time. Late arrival may result in a deduction of marks.
- Do not leave before completing the lab work assigned by the lab instructor. Even then, you must obtain permission from the instructor before leaving.
- You may talk with your group members and the lab instructor only about matters related to the lab work. Avoid gossiping, and keep your voice low so as not to disturb others.
- Do not use mobile phones to talk or for any other purpose.
- Copying lab reports is strictly prohibited.
- All lab reports must be submitted at the beginning of the class
- No late submissions will be accepted under any circumstances.

Lab Report Writing Format

- Cover page
- Objectives
- Theory
- Apparatus List
- Circuit Diagrams
- Data Table
- Graphs (If required in the experiment)
- Result Analysis
- Questions and Answers
- Discussion
- Attachment (Signed lab sheet) + Simulations

➔The Lab Report should be submitted in hand written format

➔Use Graph paper for accurate graph construction

➔Please utilize both sides of each page for the report

➔Simulation images must be submitted in a vertical orientation

Report Writing Guidelines

After completion of a lab experiment, the lab report is due in the next immediate lab class. Each group has to submit a single lab report for each experiment, and every group member must have a contribution to it. Late submissions should not be allowed. Below is a detailed description of what each lab report must contain:

1. Cover Page - All lab reports should have a cover page and the same cover page should be used for all the lab reports. A sample of a cover page will be provided to you.

2. Objectives – You should briefly write what was the aim of the experiment. In other words, write what you intend to achieve by doing the experiment.

3. Theory (write in your own words): Students should briefly explain the theory in their own words.

4. Apparatus List – A simple list of all the apparatuses and components you used to do the lab experiment.

5. Circuit Diagrams – Draw the circuit diagrams for the experiment; it should be drawn by pencil but should be clean and legible.

6. Data / Readings – This section of the lab report will contain the data that you have collected practically and it should be presented in a tabular form (make a fresh clean table of the data obtained in class).

7. Graph – Draw graphs if necessary and explain your graph in a clear and precise manner. For instance, you should be able to explain why a part of a graph rises initially and then becomes

8. Result Analysis – This is an important part of the report. You have to explain the results whether they met your objectives or not.

9. Questions and Answers: There are a few report questions that you need to answer.

10. Discussion –This is one of the most important parts of the lab report. What you write here proves how attentive and careful you were during the lab class. Copying a single line from another person's discussion or from a previous lab report will earn you a straight zero.

You must focus on these points:

- What did you learn from the experiment?
- Write whether the data and graph were exactly how you expected from the theoretical knowledge or the practical results were different from theory.
- The problems faced during the experiment and a legitimate reason for the possible fluctuation in readings, if any. You can also write about the limitations and drawbacks of the experiment. You also provide any suggestions that you deem fit.

11. Attachment– Finally, you must attach the data sheet signed by the instructor in the lab, and include simulation results with simulated circuit diagrams.

Reminder:

1. Attach the provided cover page with your lab report.

2. Write your name and ID inside the simulation as text.

3. You may use either graph paper or Excel to plot graphs.

4. Copy-pasting is strictly prohibited. Sharing reports with other groups is also not allowed. In case of any problem/confusion, consult your instructor.